less than 1 percent the efficiency of the annihilation of matter and anti-matter. But even proton-proton reactions are far beyond anything we can realistically imagine for ourselves in the near future. The required temperatures are much too high. Instead of jamming protons together, though, we might use heavier kinds of hydrogen. We already do so in thermonuclear weapons. Deuterium is a proton bound by nuclear forces to a neutron; tritium is a proton bound by nuclear forces to two neutrons. It seems likely that in another century we will have practical power schemes that involve the controlled fusion of deuterium and tritium, and of deuterium and helium. Deuterium and tritium are present as minor constituents in water (on Earth and other worlds). The kind of helium needed for fusion, ^{3}He (two protons and a neutron make up its nucleus), has been implanted over billions of years by the solar wind in the surfaces of the asteroids. These processes are not nearly as efficient as the proton-proton reactions in the Sun, but they could provide enough power to run a small city for a year from a lode of ice only a few meters in size.

Fusion reactors seem to be coming along too slowly to play a major role in solving, or even significantly mitigating, global warming. But by the twenty-second century, they ought to be widely available. With fusion rocket engines, it will be possible to move asteroids and comets around the inner Solar System, taking a main-belt asteroid, for example, and inserting it into orbit around the Earth. A world 10 kilometers across could be transported from Saturn, say, to Mars through nuclear burning of the hydrogen in an icy comet a kilometer across. (Again, I'm assuming a time of much greater political stability and safety.)

PUT ASIDE FOR THE MOMENT any qualms you might have about the ethics of rearranging worlds, or our ability to do so without catastrophic consequences. Digging out the insides of worldlets, reconfiguring them for human habitation, and moving them from one place in the Solar System to another seems to be within our grasp in another century or two. Perhaps by then we will have adequate international